Hydrogenic-demo

Hydrogenic-demo provides a compact C++ example for hydrogenic atom calculations, together with a Python interface for interactive work and notebook-based demonstrations. The C++ code builds the `hydrogenic_atom_demo` executable and exercises the routines in `Hydrogenic.vX` together with numerical utilities from `Tools`.

Project layout

Path	Purpose
`main.cpp`	Example program that constructs hydrogenic atoms and prints representative atomic data.
`Hydrogenic.vX/Atom.*`	Core `Atom`, `Atomic_Shell`, `Electron_Level`, and `Gas_of_Atoms` classes.
`Hydrogenic.vX/Oscillator_strength.*`	Bound-bound oscillator strength and transition support.
`Hydrogenic.vX/Photoionization_cross_section.*`	Photoionization cross sections and Gaunt factors.
`Hydrogenic.vX/Quadrupole_lines_HI.*`	Electric quadrupole line rates for hydrogenic atoms.
`Hydrogenic.vX/Rec_Phot_BB.*` and `Recombination_Integrals.*`	Recombination and photon interaction helpers.
`Tools/`	Shared constants, integration routines, line profiles, and simple numerical helpers.
`python/`	Installable Python package and Jupyter notebook examples.
`Documentation/`	PDF documentation and editable TeX source.
`Makefile` and `Makefile.in`	Build entry points for the C++ demo executable.

What the C++ example does

The demo constructs hydrogenic atoms with a selected number of complete shells. Each shell contains angular momentum levels with `l = 0 ... n - 1`. The code can report:

- level degeneracies, ionization energies, ionization frequencies, total decay rates, and lifetimes;
- selected bound-bound transition frequencies, wavelengths, and Einstein `A21` coefficients;
- electric dipole transitions and optional electric quadrupole transitions;
- photoionization cross sections and Gaunt factors at frequencies above threshold;
- simple rescaling behavior for the fine-structure constant and electron mass.

The current `main.cpp` includes two examples:

- hydrogen with `Z = 1`, 5 shells, quadrupole lines enabled, and Storey-Hummer recombination interpolation enabled;
- hydrogenic oxygen, O VIII, with `Z = 8`, 30 shells, quadrupole lines enabled, and Storey-Hummer recombination interpolation enabled.

Scientific background

The hydrogenic atom setup was developed for detailed calculations of cosmological recombination. This requires multi-shell hydrogenic atoms with resolved angular momentum sub-states, dipole transitions, quadrupole transitions, photoionization cross sections, and recombination coefficients. The C++ example in `main.cpp` cites the main background papers for this use case.

Dependencies

The C++ build expects:

- a C++17 compiler with OpenMP support;
- GSL headers and libraries;
- Boost headers;
- GNU `make`.

The Python package additionally uses:

- Python 3.9 or newer;
- `pybind11` and `numpy` at build time;
- optional `jupyter` and `matplotlib` for notebooks and plots.

On macOS with Homebrew, the typical compiled dependency setup is:

```
brew install gcc gsl boost
```

`Makefile.in` currently defaults to:

```
CXX = /opt/homebrew/bin/g++-16
DEF_INC_PATH = /opt/homebrew/include
DEF_LIB_PATH = /opt/homebrew/lib
```

If your compiler or Homebrew prefix differs, override these variables when invoking `make` or `pip`.

Build and run the C++ demo

From the workspace root:

```
make
```

Run the demo:

```
make run
```

Clean generated build products:

```
make clean
```

Update submodules if the project is checked out with submodule dependencies:

```
make update-modules
```

Example build override:

```
make CXX=/opt/homebrew/bin/g++-16 DEF_INC_PATH=/opt/homebrew/include
DEF_LIB_PATH=/opt/homebrew/lib
```

The executable is written to:

```
./hydrogenic_atom_demo
```

Intermediate object and dependency files are written under:

```
./build/
```

Install the Python package

From the repository root:

```
python -m pip install ./python
```

For editable development with notebook dependencies:

```
python -m pip install -e "./python[notebooks]"
```

If needed, override the compiler and dependency paths explicitly:

```
CXX=/opt/homebrew/bin/g++-16 \
GSL_INC_PATH=/opt/homebrew/include \
GSL_LIB_PATH=/opt/homebrew/lib \
BOOST_INC_PATH=/opt/homebrew/include \
python -m pip install ./python
```

Use from Python

```python
import numpy as np
from hydrogenic import hydrogen, oxygen

h = hydrogen(shells=5)
print(h.level(2, 1))
print(h.transition(2, 1, 1, 0))

o = oxygen(shells=30)
nu0 = o.level(1, 0)["ionization_frequency_hz"]
nu = np.geomspace(1.001, 10.0, 100)*nu0
data = o.photoionization(1, 0, nu)
print(data["sigma_cm2"])
```

Photoionization cross sections require `recombination_mode=1` or `recombination_mode=2`; this is the default used by the Python convenience constructors.

Python notebooks

The notebook examples live in `python/notebooks/`. They illustrate the same routines from Python, including level data, line transitions, and photoionization cross-section plots for hydrogen and hydrogenic oxygen.

Documentation files

The PDF documentation is:

`Documentation/Hydrogenic_cpp_Documentation.pdf`

The editable TeX source is:

`Documentation/Hydrogenic_cpp_Documentation.tex`

The PDF can be regenerated from this README with Pandoc and Typst:

```
pandoc README.md --from gfm --pdf-engine=typst --output Documentation/Hydrogenic_cpp_Documentation.pdf
```

If a LaTeX installation is available, the TeX source can be edited and compiled by hand:

```
cd Documentation
pdflatex Hydrogenic_cpp_Documentation.tex
```

References

[1] J. Chluba and R. M. Thomas, "Towards a complete treatment of the cosmological recombination problem", MNRAS 412, 748, 2011. https://ui.adsabs.harvard.edu/abs/2011MNRAS.412..748C/abstract

[2] J. Chluba, J. A. Rubino-Martin, and R. A. Sunyaev, "Cosmological hydrogen recombination: populations of the high level sub-states", MNRAS 374, 1310, 2007. https://ui.adsabs.harvard.edu/abs/2007MNRAS.374.1310C/abstract

[3] J. Chluba and Y. Ali-Haimoud, "CosmoSpec: Fast and detailed computation of the cosmological recombination radiation from hydrogen and helium", MNRAS 456, 3494, 2016. https://ui.adsabs.harvard.edu/abs/2016MNRAS.456.3494C/abstract